

Total No. of Questions : 10]

SEAT No. :

**P3233**

[5461]-274

[Total No. of Pages : 2

**B.E. (Computer Engineering)**

**MOBILE COMPUTING**

**(2012 Course) (Elective - III) (End Sem.) (Semester - II)**

*Time : 2½ Hours]*

*[Max. Marks : 70*

*Instructions to the candidates:*

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Assume suitable data, if necessary.

**Q1) a)** Describe mobility management with suitable diagram. **[5]**

b) Explain functioning of cellular network. How the given set of frequencies are used to increase capacity of a network? **[5]**

OR

**Q2) a)** Explain using block diagram the process of authentication in a GSM service. **[5]**

b) Explain the purpose and IEEE standards for WPAN, WLAN, Wimax, Bluetooth. **[5]**

**Q3) a)** Explain DSSS with one example. **[5]**

b) Explain the multiple access efficiency of FDMA systems. **[5]**

OR

**Q4) a)** What is handover? When handover in mobile communication takes place. **[5]**

b) Why CSMA/CD is not used in wireless networks? **[5]**

**Q5) a)** Explain working of Mobile IP. What are the limitations of traditional IP to support mobility? **[6]**

b) Explain Snooping TCP. Give the advantages and disadvantages of Snooping TCP. **[6]**

c) Explain what is MANET. What are the security threats to a MANET? **[6]**

OR

**P.T.O.**

- Q6)** a) What do you mean by tunneling, encapsulation and de-capsulation? How does a tunnel differ from a route? [6]
- b) List the entities of Mobile IP and describe data transfer from mobile node to a fixed node and vice versa. [6]
- c) What is reactive routing protocol in MANET? Describe DSR and AODV routing protocols. [6]

- Q7)** a) What are the different types of data synchronizations in mobile computing systems? Describe synchronization usage models in mobile applications. [8]
- b) Explain in detail classification of data delivery mechanism in mobile computing. [8]

OR

- Q8)** a) Explain the reason for Communication asymmetry in mobile network. Give examples of asymmetric communication architecture for data dissemination. [8]
- b) Describe Pull-based data-delivery mechanism. What are the advantages and disadvantages of Pull-based data-delivery? [8]
- Q9)** a) Describe function of mobile agent. Why does an agent move from tier to tier during an application? Distinguish between an agent and a server. [8]
- b) Explain basic characteristics of mobile operating systems and write difference between IOS and Android OS. [8]

OR

- Q10)** a) Explain the role of gateway in connecting networks using different protocols. Describe transcoding gateway and its applications in mobile computing system. [8]
- b) Write short note on [8]
- i) Service discovery and device management in mobile O.S.
- ii) Mobile-agent-based architecture.

